

Lucy Coe

✉ lucycoe07@gmail.com |  GitHub |  LinkedIn |  Portfolio | StudyFlow

Education

BSc Computer Science, University of Edinburgh

2024–2028

Predicted First Class Honours

Relevant Modules: Algorithms & Data Structures, Software Engineering and Professional Practice, Reasoning and Agents, Computer Systems.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C, SQL
Backend: FastAPI, PostgreSQL, Supabase, REST APIs, HTTP, TCP/IP
Frontend: React, PixiJS, HTML, CSS, Tailwind CSS
Testing: JUnit 5, Vitest, React Testing Library
Tools: Git, Maven, Linux (WSL), Vercel

Projects

StudyFlow

React, Supabase, PostgreSQL, Vitest

- Built and deployed a full-stack study platform across 7 PostgreSQL tables, integrating coursework, timetables, flashcards and task management.
- Engineered a constraint-aware scheduling algorithm prioritising coursework by deadline and workload; enforced per-user data isolation with Supabase Row Level Security and validated functionality with 230 automated tests.

HTTP/1.1 Server

Java, TCP Sockets, Maven, JUnit 5

- Built an HTTP/1.1 server from raw TCP sockets without a web framework, implementing parsing, dynamic routing, middleware, static files, cookies and server-side sessions.
- Implemented a 4-thread worker pool, graceful shutdown and path traversal protection; validated complete TCP request-response flows with 119 unit and integration tests.

Project Simian

Python, FastAPI, React, TypeScript, PixiJS

- Building an agent-based simulation across a 300×300 procedural world, using 32×32 chunk loading to dynamically render terrain during panning and zooming.
- Developed agents with A* pathfinding, spatial food memory and hunger, energy and mortality systems; tested simulation behaviour with 150+ concurrent agents.

Cache Simulator

C, Assembly

- Built a configurable cache simulator supporting direct, set and fully-associative caches, LRU replacement and two-level L1/L2 hierarchies while tracking cache hits and misses.

Student Event Booker

Java, JUnit

- Collaborated in a four-person Agile team to implement booking, cancellation, refund and event filtering, validated through 100+ JUnit tests.

Experience

Object-Oriented Programming Tutor and Marker

2026

University of Edinburgh

- Led weekly Java/OOP tutorials for 15+ students and assessed coursework, providing feedback on debugging, program design and code quality.

Cognitive Science Python Lab Demonstrator

2026

University of Edinburgh

- Supported weekly Python labs for 5–15 students, troubleshooting code and guiding NumPy and Jupyter-based computational exercises.

Customer Service and Hospitality Assistant

Until 2024

Café, Restaurant and Pottery Studio

- Independently managed a pottery studio during shifts, handling customer requests and processing over £300 in sales.

Awards and Achievements

Amazon Future Engineer Scholarship • Robertson Trust Scholarship • Proxime Accessit (Wick High School)